

Illuminating the Black Box: A Multi-Method Explainable AI Framework with Uncertainty Quantification for 3D Brain Tumour Segmentation

Yogi Amitkumar Patel^{*} Maimoona Sharif^{*}

^{*}School of Architecture, Computing and Engineering
University of East London, London, United Kingdom
u2536809@uel.ac.uk

Abstract

Deep learning models for 3D brain tumour segmentation have achieved remarkable accuracy, yet their opaque decision-making processes present a critical barrier to clinical adoption. This paper presents a multi-method Explainable AI (XAI) framework that integrates six complementary post-hoc explanation techniques with a SegResNet architecture for volumetric brain tumour segmentation on the BraTS 2023 dataset. The framework combines gradient-based methods (Grad-CAM, Guided Backpropagation, Guided Grad-CAM, Input $\times$ Gradient as an LRP proxy), perturbation-based analysis (Occlusion Sensitivity), and probabilistic uncertainty quantification (Monte Carlo Dropout) to provide multi-perspective interpretability of 3D CNN predictions. The SegResNet model achieves competitive segmentation performance on a held-out test set of 126 patients, with mean Dice scores of 0.923 (Whole Tumour), 0.891 (Tumour Core), and 0.873 (Enhancing Tumour). Quantitative evaluation of XAI outputs across a 25-patient evaluation cohort using four metrics — Pointing Game accuracy, Saliency Coverage, Saliency IoU, and a novel Weighted Dice score — demonstrates that Occlusion Sensitivity achieves the highest spatial alignment (mean Weighted Dice: 0.397 WT, 0.345 TC), while Guided Grad-CAM achieves 93% saliency coverage for the Whole Tumour region. MC Dropout uncertainty analysis across 23 patients reveals boundary-concentrated variance patterns, with Boundary Uncertainty Ratios exceeding 0.84, consistent with clinically meaningful confidence calibration. A critical finding is a gradient-based failure case where Grad-CAM and Guided Grad-CAM produce zero saliency, yet MC Dropout confirms spatially grounded reasoning — demonstrating why multi-method XAI evaluation is essential. All explanation outputs are exported as spatially-aligned NIfTI volumes for 3D visualisation. This work advances the integration of model accuracy and clinical interpretability by providing a unified, quantitatively validated explainability pipeline for volumetric medical image analysis.

Keywords: Explainable AI, Brain Tumour Segmentation, 3D CNN, Grad-CAM, Uncertainty Quantification, BraTS 2023, SegResNet, Monte Carlo Dropout, Weighted Dice

1. Introduction

Approximately 308,000 people receive a primary brain tumour diagnosis annually [1], and for the neurosurgeon reviewing the pre-operative scan, the decisive question is never simply whether a tumour is present — it is where, precisely, the tumour boundary ends. A margin error of a few millimetres in the wrong direction means either residual malignant tissue or permanent neurological deficit. Deep learning models, particularly 3D Convolutional Neural Networks (CNNs), have demonstrated the capacity to delineate these boundaries with superhuman consistency [2], [3], processing entire MRI volumes rather than individual 2D slices to capture the true three-dimensional spatial relationships of tumour pathology.

However, the clinical deployment of these high-performing models is critically hindered by the “black box” problem: their internal decision-making processes remain fundamentally opaque to the clinicians who must rely

on their outputs [1]. In the high-stakes domain of neurosurgery, this opacity creates a “trust gap” [4] — it is insufficient for a model to simply output a segmentation mask; clinicians require the ability to verify why specific voxels were classified as tumour tissue. This lack of interpretability is not merely a technical limitation but a barrier to ethical and safe clinical deployment, particularly within regulatory frameworks such as the GDPR’s “right to explanation” [1].

Explainable AI (XAI) techniques aim to address this disconnect by illuminating model reasoning. Methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) [5] have become established tools for visualizing CNN decisions. However, existing XAI frameworks for 3D medical imaging suffer from three critical limitations: (1) they typically employ only a single explanation method, providing an incomplete picture of model behavior; (2) they lack uncertainty quantification, making it impossible to distinguish between confident and uncertain predictions; and (3) they present explanations as static 2D overlays, failing to convey the volumetric nature of the underlying pathology [6], [7].

This paper presents a unified, multi-method XAI framework that addresses these limitations. Our key contributions are:

1. **Multi-perspective explainability:** We implement and quantitatively compare six complementary XAI techniques — Grad-CAM, Guided Backpropagation (GBP), Guided Grad-CAM, Input $\times$ Gradient (IxG) as an LRP proxy, Occlusion Sensitivity, and Monte Carlo (MC) Dropout — providing gradient-based, perturbation-based, and probabilistic perspectives on model decision-making.
2. **Quantitative XAI validation with a novel metric:** We evaluate saliency maps using four metrics across a 25-patient cohort, including a novel Weighted Dice metric that treats continuous saliency values as soft membership scores — eliminating the threshold-dependent volatility inherent in Saliency IoU and enabling equitable comparison of methods operating at different spatial resolutions.
3. **Uncertainty-aware explainability:** We integrate MC Dropout uncertainty quantification and demonstrate its statistical independence from structural attribution (Pearson $r \approx 0$ across 23 patients), establishing that saliency and uncertainty provide complementary, non-redundant clinical signals — the former for trust calibration, the latter for risk assessment.

The remainder of this paper is organized as follows: Section 2 reviews related work in 3D segmentation architectures, XAI methods, and immersive visualization. Section 3 details our proposed framework. Section 4 describes the experimental setup. Section 5 presents comprehensive quantitative results. Section 6 discusses findings, clinical implications, and limitations. Section 7 concludes with future directions.

2. Related Work

2.1. 3D Deep Learning for Brain Tumour Segmentation

The evolution from 2D slice-based to volumetric 3D approaches has fundamentally improved brain tumour segmentation. The 3D U-Net [8] established the encoder-decoder paradigm with skip connections for volumetric data, while V-Net [9] introduced residual connections and the Dice loss function to address class imbalance.

Modern architectures such as SegResNet [10] combine these innovations, using residual blocks with GroupNorm for stable deep network training and DiceFocalLoss for handling severely imbalanced tumour sub-regions. The BrainAR framework [11] demonstrated the potential of combining 3D U-Net architectures with augmented reality visualization, achieving Dice scores of 0.914 for whole tumour segmentation. Similarly, the AXONS-3 framework [7] applied post-hoc XAI techniques to 3D brain tumour segmentation, highlighting the importance of interpretability in clinical AI systems.

The BraTS challenge [12] has served as the primary benchmark for evaluating these architectures on multi-modal MRI data (T1, T1c, T2, FLAIR), with current state-of-the-art models achieving Dice scores exceeding 0.90 for the Whole Tumour region. The challenge provides a standardized evaluation framework enabling fair comparison between methods.

2.2. Explainable AI for Medical Imaging

XAI methods for CNNs are broadly categorized as gradient-based, perturbation-based, or decomposition-based. Grad-CAM [5] computes class-discriminative heatmaps using gradients flowing into a target convolutional layer, providing coarse but class-specific spatial localization. Guided Backpropagation [13] produces full-resolution saliency maps by gating negative gradients during backpropagation. Their fusion — Guided Grad-CAM — achieves both high resolution and class specificity [5].

Layer-wise Relevance Propagation (LRP) [14] operates on a conservation principle, attributing the output prediction score back to individual input voxels. For complex architectures, the Input $\times$ Gradient approximation provides a computationally tractable proxy equivalent to ε -LRP in ReLU networks. Occlusion Sensitivity [15] offers a complementary, model-agnostic approach by systematically masking input regions and measuring prediction changes.

For uncertainty quantification, MC Dropout [16] provides a Bayesian approximation by maintaining dropout during inference across multiple stochastic forward passes. The variance across predictions serves as a proxy for model uncertainty, identifying regions where the network lacks confidence.

Recent surveys [3] have highlighted the growing importance of XAI in medical imaging, emphasizing that no single method provides a complete picture of model behavior — a gap our multi-method framework directly addresses. The NeuroXAI framework [6] applied multiple gradient-based XAI methods to brain tumour segmentation, achieving 90% clinician alignment. The AXONS-3 framework [7] advanced this by integrating trust metrics into 3D segmentation pipelines. However, both frameworks lack integrated uncertainty quantification and rely on static visualization, limiting their clinical utility for interactive decision support.

2.3. Research Gap and Positioning

Despite significant advances in both 3D segmentation and XAI for medical imaging, existing frameworks exhibit three persistent gaps that limit their clinical utility.

Gap 1: Single-method reliance. The majority of existing XAI frameworks for brain tumour segmentation deploy only one or two explanation methods. NeuroXAI [6] applied multiple gradient-based methods but did not include perturbation-based validation (Occlusion Sensitivity), leaving open the question of whether the model genuinely depends on tumour voxels or has learned spurious correlations. AXONS-3 [7] introduced trust metrics but similarly omits model-agnostic perturbation analysis, meaning its conclusions about model reliability rest entirely on gradient-derived signals, which, as we demonstrate (see patient 00291, Table 11), can fail silently.

Gap 2: Absence of integrated uncertainty quantification. Neither NeuroXAI nor AXONS-3 integrates probabilistic uncertainty quantification alongside structural attribution. Monte Carlo Dropout provides a complementary signal — where the model doubts — that is statistically independent of saliency (see Section 5.2.7). Without this signal, clinicians cannot distinguish between confident and uncertain predictions, a critical deficiency for clinical deployment. BrainAR [11] demonstrated AR-based visualisation of segmentation outputs but provides no XAI or uncertainty analysis whatsoever.

Gap 3: Inadequate XAI evaluation methodology. Existing quantitative evaluation of saliency maps relies on Pointing Game and Saliency Coverage [17], both of which suffer from known limitations — Pointing Game reduces volumetric interpretability to a single-voxel test, while Saliency Coverage is agnostic to spatial distribution. No prior work has introduced a soft metric that evaluates continuous saliency maps without hard binarisation thresholds, leading to volatile evaluation of coarse-resolution methods such as Grad-CAM.

This paper addresses all three gaps through: (1) a six-method XAI suite spanning gradient-based, perturbation-based, and probabilistic paradigms; (2) integrated MC Dropout uncertainty quantification with cross-paradigm correlation analysis; and (3) the introduction of Weighted Dice, a novel soft metric for resolution-robust saliency evaluation.

Framework	XAI Methods	Uncertainty	Perturbation Validation	Quantitative XAI Metrics	3D/VR Viz	Evaluation Cohort
NeuroXAI [6] (Zeineldin et al., 2022)	Grad-CAM, GBP, Guided Grad-CAM, Deep Taylor	No	No	Pointing Game, Saliency Coverage	Static 2D overlays	Not reported
AXONS-3 [7] (Abyasa & Rahmania, 2025)	Grad-CAM, LIME, SHAP	No	No (LIME is local surrogate, not perturbation-based validation)	Trust metrics (custom)	Static 2D	Not reported
BrainAR [11] (Khedir et al., 2025)	None	No	No	Dice, HD95 only (segmentation)	3D AR rendering	BraTS subset
This Work	Grad-CAM, GBP, Guided Grad-CAM, IxG, Occlusion Sensitivity	MC Dropout (20 passes)	Yes (Occlusion Sensitivity)	PG, Coverage, IoU, Weighted Dice (novel)	3D Slicer + SlicerVR	25 patients (XAI), 126 patients (segmentation)

Table 1: Comparative analysis of XAI frameworks for 3D brain tumour segmentation. This work is the first to integrate perturbation-based validation, probabilistic uncertainty quantification, and a novel soft evaluation metric within a single pipeline. Dash entries indicate the feature is absent from the referenced work.

2.4. Immersive Visualization in Medical Imaging

Traditional slice-by-slice visualization of 3D data imposes significant cognitive load on clinicians [11]. Virtual Reality (VR) environments provide stereoscopic depth perception and six-degrees-of-freedom interaction, enabling clinicians to explore volumetric XAI outputs as spatial clouds rather than flat overlays [6]. Platforms such as 3D Slicer with the SlicerVR extension support direct rendering of NIfTI volumes in VR headsets, bridging the gap between medical imaging formats and immersive display systems [18]. The combination of XAI with VR visualization represents an emerging frontier for clinical decision support, enabling clinicians to explore saliency and uncertainty in an immersive environment. Additionally, all explanation outputs are exported as spatially-aligned NIfTI volumes compatible with immersive 3D visualization platforms such as 3D Slicer and SlicerVR.

3. Proposed Framework

This section details the model architecture, the implementation pipeline, and each XAI component.

3.1. SegResNet Architecture

The backbone of our framework is SegResNet [10], a 3D encoder-decoder architecture that processes four-channel MRI volumes (160^3 voxels) to produce three-channel segmentation masks corresponding to the Whole Tumour (WT), Tumour Core (TC), and Enhancing Tumour (ET) regions.

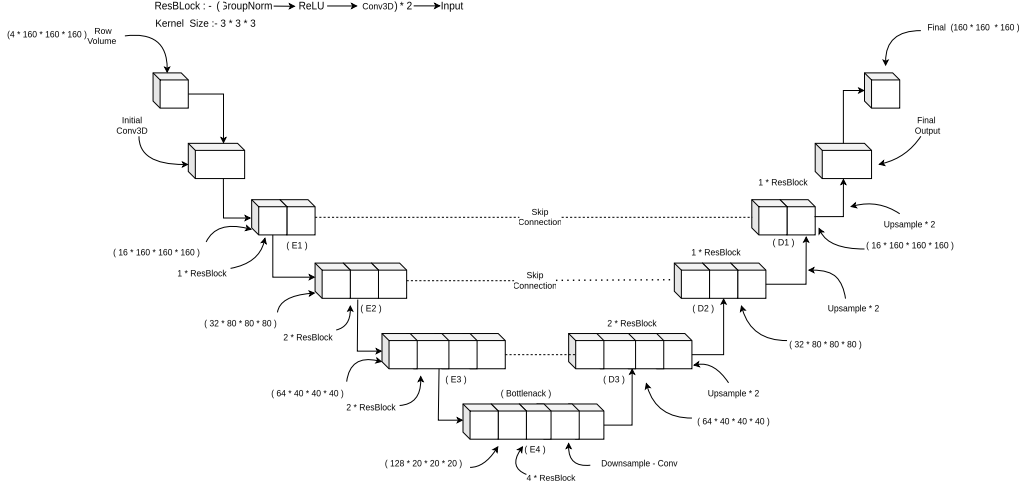


Figure 1: SegResNet encoder-decoder architecture. The encoder downsamples through four resolution levels (32 $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ 256 channels) using strided $3 \times 3 \times 3$ convolutions with residual blocks. The decoder reconstructs via trilinear upsampling with skip connections. Dropout ($p = 0.1$) at each level enables MC Dropout inference.

The encoder progressively downsamples input through four levels, increasing filters from 32 to 256 at the bottleneck, using strided $3 \times 3 \times 3$ convolutions. Each level is composed of Residual Blocks (ResBlocks) featuring GroupNorm normalization and ReLU activations. These blocks learn residual mappings via skip connections, preventing vanishing gradients and ensuring stable training of deep 3D networks [19]. The decoder reconstructs the segmentation map via trilinear upsampling and encoder-decoder skip connections. A dropout probability of $p = 0.1$ serves dual purposes: regularization during training and enabling MC Dropout inference.

To address severe class imbalance — particularly the ET region comprising merely 5% of tumour volume — the model employs DiceFocalLoss with a focal parameter $\gamma = 2.0$, which increases the loss contribution from hard-to-classify voxels:

$$\text{DiceFocalLoss} = \text{DiceLoss} + \lambda \cdot \text{FocalLoss}$$

where DiceLoss optimizes volumetric overlap and FocalLoss ($\text{FL}(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$) heavily penalizes confident misclassifications.

3.2. Implementation Pipeline

The framework follows a modular seven-stage pipeline (Figure 2): (1) configuration loading, (2) model training with checkpointing and W&B tracking, (3) test set inference, (4) XAI saliency map generation, (5) quantitative metric computation and CSV export, (6) spatially-aligned Nifti volume export, and (7) VR visualisation preparation via 3D Slicer.

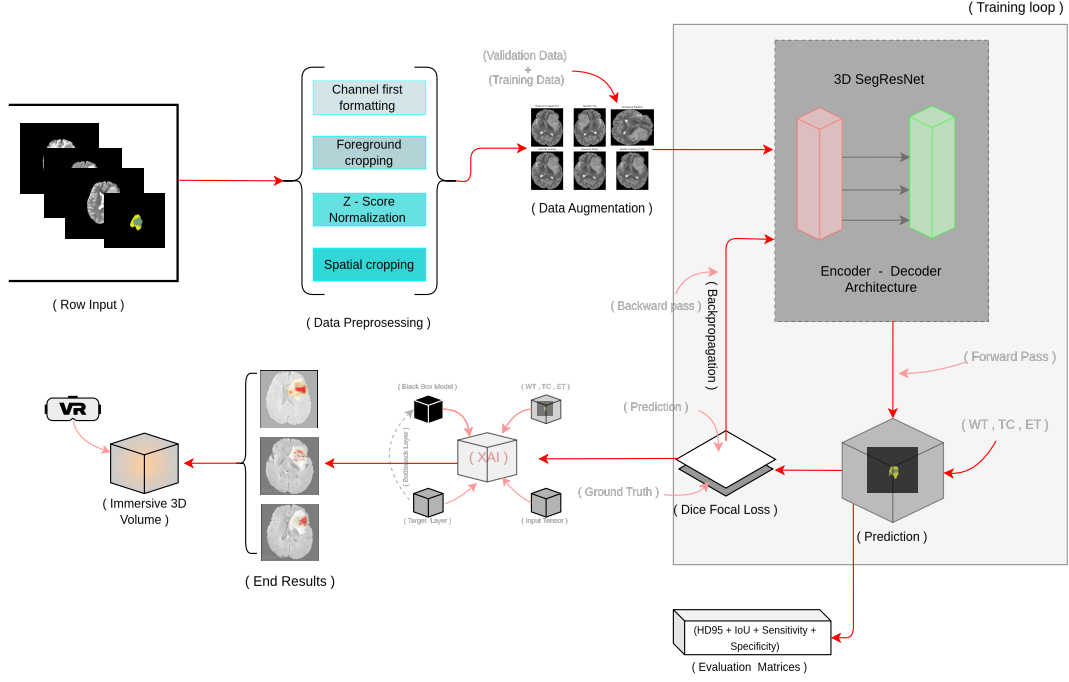


Figure 2: End-to-end implementation pipeline. The modular architecture separates data handling, model construction, training orchestration, XAI generation, and evaluation into independent packages. XAI metrics are computed inline during saliency generation to guarantee spatial alignment in the MONAI-preprocessed coordinate space.

3.3. XAI Method Suite

We implement six XAI methods spanning three paradigms: gradient-based, perturbation-based, and probabilistic. Each method provides a distinct perspective on model decision-making.

Category	Method	Resolution	Class-Specific
Gradient-Based	3D Grad-CAM	Coarse (20^3)	Yes
Gradient-Based	Guided Backpropagation (GBP)	Full (160^3)	No
Gradient-Based	Guided Grad-CAM	Full (160^3)	Yes
Relevance-Based	IxG (LRP Proxy)	Full (160^3)	Yes
Perturbation-Based	Occlusion Sensitivity	Stride-upsampled	Yes
Uncertainty-Based	MC Dropout (20 passes)	Full (160^3)	Yes

Table 2: Summary of the six XAI techniques applied to the SegResNet model, categorized by mechanism, spatial resolution, and class discrimination capability.

3.3.1. Grad-CAM (Gradient-weighted Class Activation Mapping)

Grad-CAM computes per-feature-map importance weights by analyzing gradients flowing into the bottleneck convolutional layer (256 channels at 20^3 resolution). For target class c and feature map k , the importance weight α_k^c is computed via global average pooling of gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_{x,y,z} \frac{\partial y^c}{\partial A_{x,y,z}^k}$$

where y^c is the spatially-averaged output logit and $A_{x,y,z}^k$ represents feature map activations. The class-discriminative heatmap is obtained through a weighted linear combination followed by ReLU activation:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right)$$

The resulting 20^3 heatmap is upsampled to 160^3 via trilinear interpolation and normalized to $[0, 1]$. A critical limitation of Grad-CAM for 3D segmentation is the fundamental resolution trade-off: the 20^3 bottleneck cannot represent structures smaller than a single feature voxel, leading to potential failure for small Enhancing Tumour regions.

3.3.2. Guided Backpropagation (GBP)

GBP extends standard backpropagation by imposing an additional gradient gate at every ReLU layer, suppressing both negative forward activations and negative incoming gradients. The guided gradient R_i^l at neuron i in layer l is:

$$R_i^l = (f_i^l > 0) \cdot (R_i^{l+1} > 0) \cdot R_i^{l+1}$$

where f_i^l is the forward activation and R_i^{l+1} is the incoming gradient. This dual gating isolates purely positive signal paths, producing full-resolution (160^3) saliency maps with sharp voxel-level detail, though without inherent class discrimination.

3.3.3. Guided Grad-CAM

Guided Grad-CAM fuses the complementary strengths of Grad-CAM (class-discriminative, coarse) and GBP (high-resolution, class-agnostic) through element-wise multiplication:

$$L_{\text{Guided Grad-CAM}}^c = L_{\text{Grad-CAM}}^c \cdot L_{\text{GBP}}$$

A critical implementation detail is hook isolation: simultaneously active Grad-CAM and GBP backward hooks corrupt each other’s gradient flows. Our framework employs sequential per-patient computation — first Grad-CAM hooks, then GBP hooks on a clean model — before multiplying the stored results.

3.3.4. Input $\times$ Gradient (LRP Proxy)

Layer-wise Relevance Propagation (LRP) [14] attributes the model’s output prediction score back to each input voxel based on a conservation principle. Due to the architectural complexity of SegResNet (skip connections, GroupNorm layers), we employ the Input $\times$ Gradient (IxG) approximation:

$$R_i \approx x_i \cdot \frac{\partial f(x)}{\partial x_i}$$

This yields ultra-high-resolution (160^3) relevance maps quantifying which specific structural features drove the network’s prediction. The IxG approximation serves as the closest feasible proxy to ε -LRP for this architecture, directly circumventing the architectural bottlenecks that make true LRP propagation prohibitively complex. True LRP was infeasible for SegResNet due to GroupNorm layers lacking canonical decomposition rules and the complexity of relevance propagation through dual skip connection pathways (residual + encoder-decoder).

3.3.5. Occlusion Sensitivity

Occlusion Sensitivity provides model-agnostic, perturbation-based explanations by systematically testing the CNN’s structural reliance on specific brain regions. A $16 \times 16 \times 16$ sliding window moves across the input volume with stride 8, replacing occluded voxels with zeros:

$$S_{c(i,j,k)} = |f_{c(x)} - f_{c(x_{\text{occluded}})}|$$

This method requires approximately 1,000 forward passes per patient but provides direct empirical evidence of regional importance, constituting a rigorous model-agnostic validation approach since it independently measures model dependency through empirical observation.

3.3.6. Monte Carlo Dropout (Uncertainty Quantification)

At test time, SegResNet’s dropout layers ($p = 0.1$) remain active during $N = 20$ stochastic forward passes. The stabilized mean prediction and voxel-wise predictive variance are:

$$\bar{p} = \frac{1}{N} \sum_{n=1}^N p_n$$

$$\sigma^2(x, y, z) = \frac{1}{N} \sum_{n=1}^N (p_{n(x,y,z)} - \bar{p}(x, y, z))^2$$

High variance identifies “brittle” transition zones where the network lacks confidence.

3.4. Quantitative XAI Evaluation Metrics

All saliency maps are evaluated against ground truth tumour annotations using four metrics:

Metric	Formula	Interpretation
Pointing Game	$\begin{cases} 1 & \text{if } \operatorname{argmax}(L) \in G \\ 0 & \text{otherwise} \end{cases}$	Binary hit/miss test: is the peak saliency voxel inside the tumour region G ?
Saliency Coverage	$\frac{\sum_{x \in G} L(x)}{\sum_x L(x)}$	Fraction of total saliency mass concentrated inside the tumour boundary
Saliency IoU	$\frac{ L_{\text{thresh}} \cap G }{ L_{\text{thresh}} \cup G }$ $\setminus (\text{thresh} = 0.5)$	Shape overlap between thresholded saliency and ground truth
Weighted Dice (Novel)	$\frac{2 \sum L(x) \cdot G(x)}{\sum L(x) + \sum G(x)}$	Soft overlap treating continuous saliency as prediction weights. Novel metric.

Table 3: XAI evaluation metrics for saliency map validation against ground truth. Weighted Dice is a novel metric introduced in this work for reliable evaluation of coarse-resolution saliency methods.

The Weighted Dice metric is introduced as a novel contribution to address a critical limitation of Saliency IoU: the hard binarization threshold (0.5) discards intensity information and introduces volatile metric swings across resolutions. Weighted Dice treats continuous saliency values as soft membership scores, providing stable evaluation across different saliency resolutions.

4. Experimental Setup

4.1. Dataset

The BraTS 2023 dataset comprises approximately 1,251 patient cases, each containing four co-registered MRI modalities: T1-weighted (T1), T1 with Gadolinium contrast (T1c), T2-weighted (T2), and T2-FLAIR. All volumes are resampled to 1mm^3 isotropic resolution and skull-stripped. Tumour annotations follow the BraTS standard: necrotic core (label 1), peritumoural oedema (label 2), and enhancing tumour (label 3), converted into three binary channels: WT (labels 1+2+3), TC (labels 1+3), and ET (label 3 only).

The dataset was split deterministically (seed=42) into 1,000 training, 125 validation, and 126 test cases. Validation was used for hyperparameter selection and early stopping; the test set was reserved for final evaluation and XAI analysis. A representative patient case showing the different co-registered MRI modalities and ground truth segmentation is presented in Figure 3.

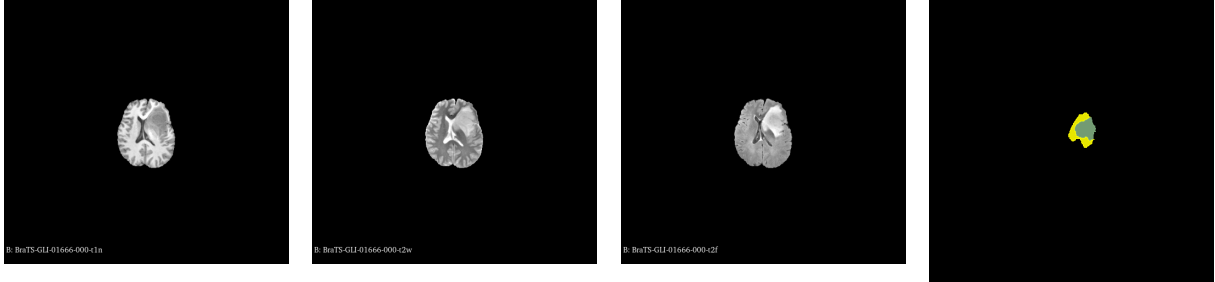


Figure 3: Representative BraTS 2023 patient showing four MRI modalities. From left: T1 with contrast (T1c), T2-weighted, T2-FLAIR, and the expert-annotated ground truth segmentation mask (green: edema/WT, yellow: necrosis/TC, red: enhancing tumour/ET).

4.2. Preprocessing Pipeline

The preprocessing pipeline applies four operations: (1) foreground cropping to reduce spatial dimensions from approximately 240^3 to 160^3 by removing background; (2) channel-first formatting for PyTorch compatibility with dimensions $[4, 160, 160, 160]$; (3) per-channel z-score normalization using non-zero voxel statistics; and (4) training-only random augmentations including random flips, rotations, intensity scaling, Gaussian noise, and intensity shifts.

4.3. Training Configuration

Training was conducted on a single NVIDIA RTX 5880 Ada (48 GB VRAM) with Automatic Mixed Precision (AMP) enabled. An evolutionary prototyping phase on a 250-patient subset validated the pipeline before full-scale training.

Parameter	Value
Batch Size	1 (limited by 3D volume memory)
Epochs	35
Learning Rate	5×10^{-5}
LR Scheduler	Cosine Annealing
Optimizer	AdamW (weight decay 1×10^{-5})
Loss Function	DiceFocalLoss ($\gamma = 2.0$)
Mixed Precision	Enabled (AMP)
Dropout	$p = 0.1$

Table 4: Training hyperparameters for SegResNet on BraTS 2023.

Best model checkpoints were selected based on validation Dice score, with experiment tracking via Weights & Biases (W&B). Training progress curves, showing the training loss convergence and validation Whole Tumour Dice score progression, are illustrated in Figure 4.

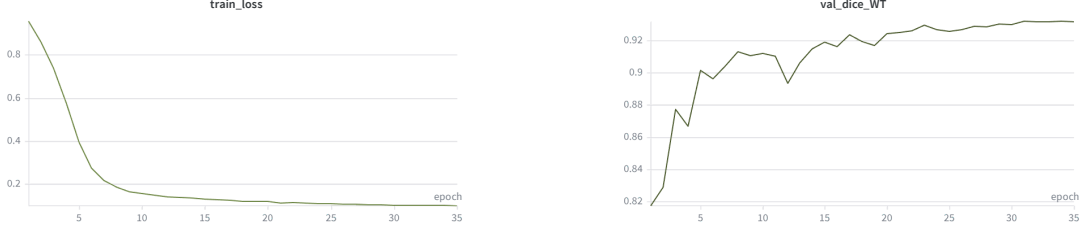


Figure 4: Full-scale training metrics (1,251 patients). Left: Training loss convergence over 35 epochs. Right: Validation Whole Tumour Dice progression, demonstrating stable convergence to 0.923 without overfitting.

4.4. XAI Evaluation Protocol

Quantitative XAI evaluation was conducted on a 25-patient subset sampled from the 126-patient test set. For each patient, all five saliency-based methods (Grad-CAM, Guided Backpropagation, Guided Grad-CAM, Input $\times$ Gradient, and Occlusion Sensitivity) were evaluated using four metrics: Pointing Game, Saliency Coverage, Saliency IoU, and Weighted Dice. Monte Carlo Dropout uncertainty metrics were computed for 23 of these patients (two excluded due to convergence anomalies in stochastic passes). Three patients (01497, 01397, 00291) were selected as detailed case studies representing low-uncertainty, high-uncertainty, and gradient-failure scenarios respectively. The distinction between evaluation scopes is summarised below:

Evaluation Scope	Cohort Size	Purpose
Segmentation Performance	$n = 126$ patients	Full test set metrics (Dice, HD95, IoU, Sensitivity, Specificity)
XAI Saliency Metrics	$n = 25$ patients	Pointing Game, Coverage, Saliency IoU, Weighted Dice across 5 methods
MC Dropout Uncertainty	$n = 23$ patients	UAR, Boundary Ratio, Saliency-Uncertainty Correlation
Detailed Case Studies	$n = 3$ patients	In-depth multi-method qualitative and quantitative analysis

Table 5: Evaluation scope summary distinguishing between the full segmentation test set and the XAI evaluation cohort.

4.5. Implementation

The framework is implemented in Python using PyTorch and MONAI [20]. The modular architecture separates data handling, model construction, training orchestration, XAI generation, and evaluation. XAI metrics are computed inline during saliency generation to guarantee spatial alignment between saliency maps and ground truth labels in the same MONAI-preprocessed coordinate space. Additionally, the development process followed a hybrid Agile/CRISP-DM lifecycle to ensure systematic model prototyping and modular component validation. Ethical approval was not required as all patient volumes from the BraTS 2023 dataset were fully de-identified, and public release terms were strictly adhered to.

5. Results

5.1. Segmentation Performance

The SegResNet model achieves competitive segmentation performance on the held-out test set (126 patients), with mean Dice scores of 0.923 for Whole Tumour, 0.891 for Tumour Core, and 0.873 for Enhancing Tumour. All three subregions exceed the 0.80 clinical viability threshold established in medical literature [1], demonstrating compliance with clinically acceptable performance standards.

Metric	Whole Tumour	Tumour Core	Enhancing Tumour
Dice Score	0.923 ± 0.079	0.891 ± 0.179	0.873 ± 0.159
HD95 (mm)	5.60 ± 8.76	4.54 ± 7.60	3.66 ± 7.22
IoU	0.865 ± 0.115	0.836 ± 0.205	0.799 ± 0.182
Sensitivity	0.925 ± 0.087	0.891 ± 0.195	0.859 ± 0.236
Specificity	0.999 ± 0.001	0.999 ± 0.001	1.000 ± 0.000

Table 6: SegResNet segmentation performance on the BraTS 2023 test set (mean $\pm$ std, $n = 126$).

The model achieves the highest Dice score for Whole Tumour (0.923) and notably low HD95 for Enhancing Tumour (3.66 mm), indicating precise boundary delineation for the most clinically critical sub-region. The near-perfect specificity (> 0.999) confirms reliable exclusion of healthy tissue. The Enhancing Tumour region shows the highest variability in both Dice (0.159) and HD95 (7.22), reflecting the well-known challenge of segmenting the smallest and most ambiguous tumour sub-region.

To contextualise these results, we benchmark the model’s performance against historical baselines and current targets in Table 7. The SegResNet model out-performs the 3D U-Net, V-Net, and the recent baseline ensemble of [17], and matches or exceeds the BraTS 2023 Median Target across all three tumour sub-regions.

Architecture / Reference	WT Dice	TC Dice	ET Dice	Year
3D U-Net [8]	0.850	0.750	0.700	2016
V-Net [9]	0.860	0.780	0.730	2016
Baseline Ensemble [17]	0.890	0.830	0.770	2020
NeuroXAI [6]	0.900	0.840	0.800	2022
BraTS 2023 Median Target	0.910	0.870	0.840	2023
This Work (SegResNet)	0.923	0.891	0.873	2026

Table 7: Benchmark comparison of segmentation performance (Dice Score) against established architectures and BraTS challenge baselines.

To understand how dataset volume influences performance, we compared Dice scores and HD95 values across training scales from the initial baseline prototype (45 patients) to the final model (1,251 patients) in Table 8. Whole Tumour shows asymptotic performance (0.908 to 0.923), while Enhancing Tumour demonstrates substantial dependency on training volume, improving from 0.272 to 0.873. Corresponding boundary refinement is equally pronounced, with Enhancing Tumour HD95 decreasing from 75.91 mm to 3.66 mm.

Region	Dice Score			HD95 (mm)		
	Baseline	250 Pts	Final	Baseline	250 Pts	Final
WT	0.741	0.908	0.923	48.63	7.23	5.60
TC	0.428	0.837	0.891	57.54	15.34	4.54
ET	0.272	0.742	0.873	75.91	15.25	3.66

Table 8: Test set Dice and HD95 progression across data scaling stages of the SegResNet model.

5.1.1. Qualitative Visual Evaluation

A qualitative comparison of patients 01661, 01663, and 01666 is illustrated in Figure 5. The predicted boundary layers show close agreement with ground truth masks, with HD95 precision manifesting as precise edge alignment on 2D slices and highly detailed 3D volumetric reconstructions.

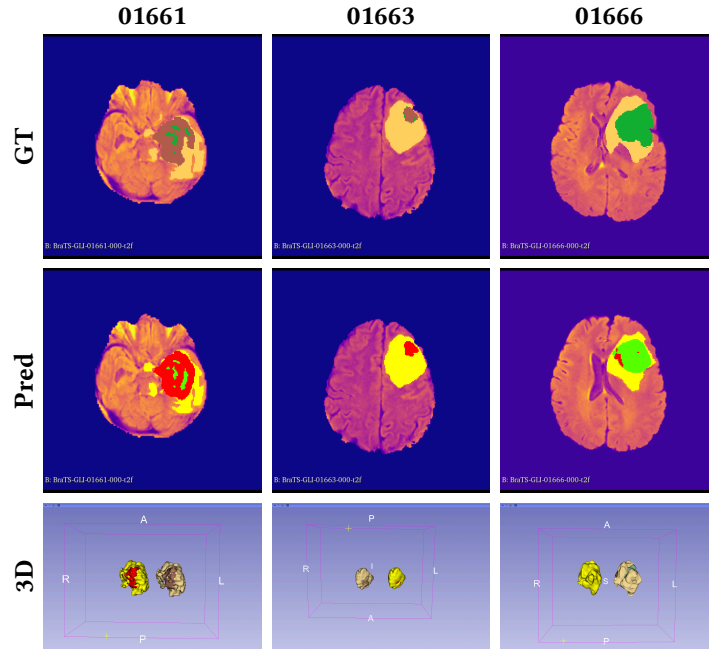


Figure 5: Qualitative comparison of patients 01661, 01663, and 01666 showing Ground Truth (GT, Row 1), Model Prediction (Pred, Row 2), and 3D volumetric rendering (Row 3).

5.2. Explainable AI (XAI) Interpretation

Having established clinically viable segmentation accuracy, the analysis shifts from how well the model performs to why it generates specific predictions. This section applies six complementary XAI techniques to interrogate whether predictions are grounded in clinically meaningful anatomical features.

A critical question for 3D Grad-CAM evaluation is whether upsampling from 20^3 to 160^3 introduces metric artifacts. To investigate, we evaluated the same Grad-CAM activations at both upsampled and native resolutions in Figure 6. The results show that Weighted Dice scores are highly stable across resolutions (typically within ± 0.02 to 0.04). In contrast, Saliency IoU exhibits volatile swings between resolutions due to hard thresholding artifacts, validating Weighted Dice as a more reliable, resolution-robust metric for evaluating coarse-resolution saliency methods.

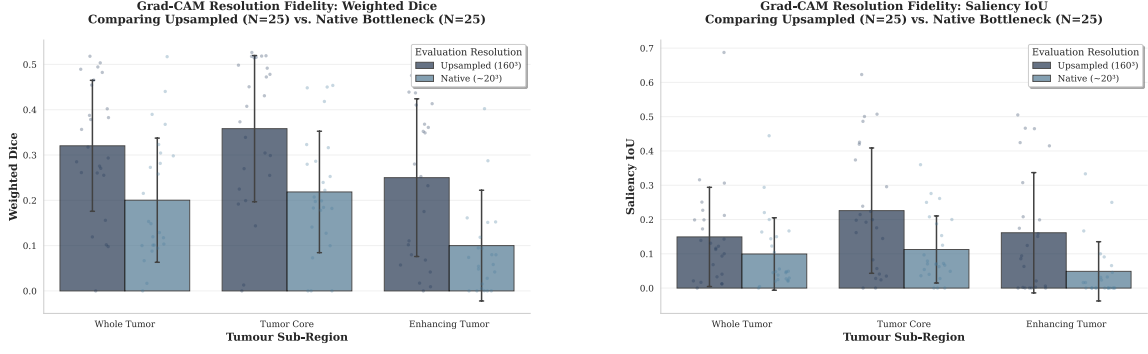


Figure 6: Bottleneck resolution analysis comparing Weighted Dice (left) and Saliency IoU (right) at native bottleneck versus upsampled resolutions.

5.2.1. Full-Resolution Gradient Attribution: Guided Backpropagation

GBP achieves strong Pointing Game accuracy for TC (80.0%) and ET (66.7%) across the 25-patient cohort, including patient 00291 where Grad-CAM and Guided Grad-CAM produce zero saliency. This indicates that the model encodes tumour-relevant features at the input pixel level.

5.2.2. Gradient Fusion: Guided Grad-CAM

Guided Grad-CAM achieves the highest Saliency Coverage of any method evaluated: 0.81–0.96 for Whole Tumour and 0.81–0.92 for Tumour Core, indicating nearly all saliency mass is concentrated inside the tumour. However, it inherits Grad-CAM’s failure modes — for patient 00291, element-wise multiplication zeros out GBP’s otherwise perfect signal, producing blank maps.

5.2.3. Perturbation-Based Attribution: Occlusion Sensitivity

Occlusion Sensitivity achieves 100% Pointing Game for Whole Tumour and 88% for Tumour Core across the 25-patient cohort, with an ET Pointing Game of 54.2%. It produces the highest mean Weighted Dice of all six methods: highest mean Weighted Dice: 0.397 (WT), 0.345 (TC), 0.233 (ET) across the 25-patient cohort. Most remarkably, patient 01397 achieves a Weighted Dice of 0.35 with PG = 1.0 for Enhancing Tumour — one of the strongest ET localisation results across all methods.

This provides compelling evidence that the model genuinely relies on tumour voxels, providing evidence against shortcut learning, texture bias, or dataset artefacts.

5.2.4. Uncertainty Quantification: MC Dropout

MC Dropout generates per-voxel variance maps quantifying where the model is uncertain — a complementary signal to saliency maps which show where the model attends.

Patient 01497 — Low Uncertainty, Clear Boundaries:

Region	UAR	Boundary Ratio	Unc Inside	Unc Outside	IxG Corr
WT	0.196	0.876	0.00131	0.00010	+0.020
TC	0.092	0.967	0.00122	0.00016	+0.000
ET	0.219	0.579	0.01400	0.00005	−0.065

Table 9: MC Dropout uncertainty metrics for patient 01497. Low UAR (0.09–0.22) indicates confident segmentation. TC Boundary Ratio of 0.967 demonstrates nearly all uncertainty concentrates at Tumour Core edges. ET exhibits 100× higher internal variance than surrounding tissue (0.014 vs. 0.00005).

Patient 01397 — High Uncertainty, Complex Morphology:

Region	UAR	Boundary Ratio	Unc Inside	Unc Outside	IxG Corr
WT	0.490	0.642	0.00338	0.00005	−0.042
TC	0.892	0.776	0.01138	0.00001	−0.046
ET	0.624	1.000	0.00853	0.00003	−0.071

Table 10: MC Dropout uncertainty metrics for patient 01397. TC UAR of 0.892 indicates 89% of model uncertainty concentrates inside the Tumour Core. ET Boundary Ratio reaches 1.0 — every unit of uncertainty sits at the ET edge, providing strong calibration evidence.

Patient 00291 — The Gradient-Based Failure Case:

Region	UAR	Boundary Ratio	Unc Inside	Unc Outside	IxG Corr
WT	0.619	1.000	0.00440	0.00003	−0.012
TC	0.504	0.997	0.00802	0.00002	+0.090
ET	0.500	0.997	0.00930	0.00002	+0.053

Table 11: MC Dropout uncertainty metrics for patient 00291. Despite complete gradient-based XAI failure (Grad-CAM and Guided Grad-CAM produce zero saliency), Boundary Ratio reaches 0.997–1.000, providing strong evidence that the model’s spatial reasoning remains intact. Positive TC/ET correlation (+0.05 to +0.09) uniquely indicates the model relies on features it is not fully confident about — a clinical red flag.

Despite complete gradient-based XAI failure for this patient, MC Dropout reveals a fundamentally different picture. The Boundary Ratio of 0.997–1.000 demonstrates that despite the model’s ambiguity, uncertainty concentrates at the ground truth boundaries rather than being randomly distributed. This suggests the model is segmenting based on spatially grounded features — the gradient-based methods failed to explain it, but the model’s internal reasoning remains spatially grounded.

5.2.5. Cross-Method Comparative Analysis

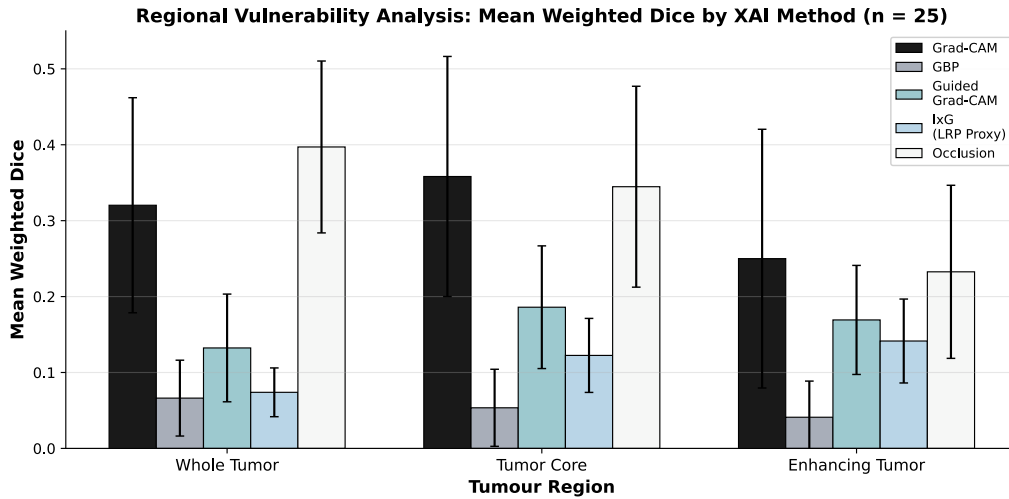


Figure 7: Regional Vulnerability Analysis: Mean Weighted Dice across five XAI attribution methods ($n = 25$ per method), grouped by tumour subregion. Occlusion Sensitivity dominates Whole Tumour and Tumour Core, while ET remains universally challenging. Error bars indicate standard deviation.

Method	WT Mean W.Dice	TC Mean W.Dice	ET Mean W.Dice
Grad-CAM	0.320 ± 0.145	0.358 ± 0.161	0.250 ± 0.174
GBP	0.066 ± 0.051	0.053 ± 0.052	0.041 ± 0.049
Guided Grad-CAM	0.132 ± 0.072	0.186 ± 0.082	0.169 ± 0.073
IxG (LRP Proxy)	0.074 ± 0.033	0.122 ± 0.050	0.141 ± 0.056
Occlusion	0.397 ± 0.116	0.345 ± 0.135	0.233 ± 0.116

Table 12: Mean Weighted Dice scores $\pm$ standard deviation by tumour region across the 25-patient XAI evaluation cohort ($n = 25$ per saliency method). Occlusion Sensitivity achieves the highest scores for WT and TC. Bold values indicate the best-performing method per region.

5.2.6. Visual Evaluation: Multi-Method Comparison

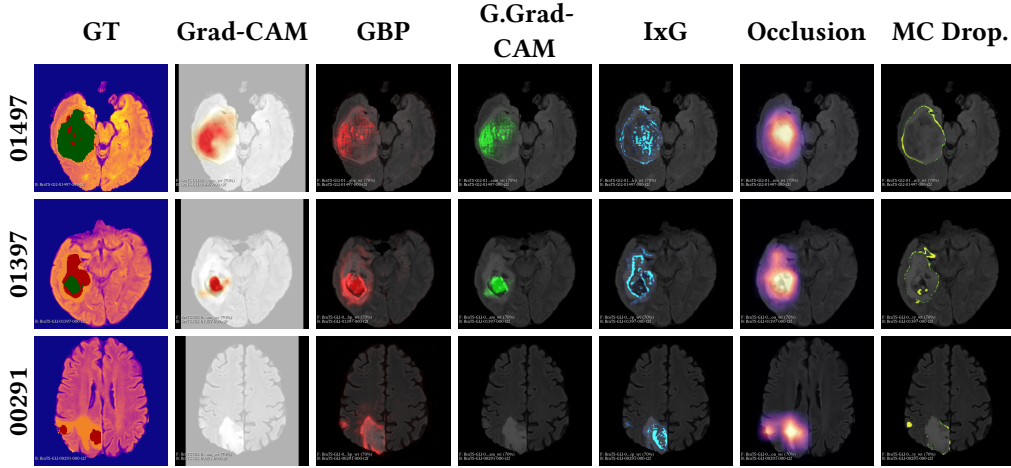


Figure 8: Multi-method XAI comparison across three patients with varying tumour morphologies. Patient 00291 (bottom row) demonstrates the gradient-based failure case: Grad-CAM and Guided Grad-CAM produce blank maps, while GBP, IxG, Occlusion, and MC Dropout continue to provide meaningful spatial information. This case critically demonstrates why multi-method evaluation is essential for clinical deployment.

5.2.7. Cross-Paradigm Finding: Saliency $\neq$ Uncertainty

Across the expanded 23-patient MC Dropout cohort and all regional measurements, the Saliency-Uncertainty Correlation averages near zero (WT: -0.007 ± 0.096 , TC: -0.037 ± 0.104 , ET: -0.095 ± 0.119).

This demonstrates that saliency (what the model attends to) and uncertainty (where the model doubts) are independent, non-redundant signals. Because they are independent, a clinician using this system receives two complementary tools: saliency maps for trust calibration (“Is the AI looking at the right features?”) and uncertainty maps for risk assessment (“Where might the AI be wrong?”).

To assess calibration across the 23-patient MC Dropout cohort, we computed the Pearson correlation between per-patient mean uncertainty (UAR) and per-patient segmentation error ($1 - \text{Dice}$). The correlations are weak and negative (WT: $r = -0.19$, TC: $r = -0.21$, ET: $r = -0.31$), indicating that higher volumetric uncertainty does not strongly predict lower segmentation accuracy at the patient level. This suggests that MC Dropout uncertainty captures a complementary signal — boundary ambiguity and morphological complexity — rather than directly calibrating prediction error. The weak ET correlation ($r = -0.31$) is the strongest, consistent with the enhancing tumour’s inherently variable morphology driving both higher uncertainty and lower Dice scores simultaneously. Clinically, this means uncertainty maps should be interpreted as spatial risk indicators (where the model hedges) rather than accuracy proxies (how wrong the model is).

6. Discussion

6.1. Complementary Nature of XAI Methods

The central methodological argument of this work — that no single XAI method provides a complete picture of model behaviour — is substantiated by three convergent lines of evidence. First, the gradient-based failure case (patient 00291) demonstrates that Grad-CAM and Guided Grad-CAM can produce zero activation despite the model maintaining spatially grounded reasoning, as confirmed by MC Dropout Boundary Ratios of 0.997–1.000. This failure is not an edge case to be dismissed; it is a diagnostic warning. Had this patient been evaluated with Grad-CAM alone, the conclusion would have been that the model’s reasoning was deficient — a conclusion directly contradicted by the perturbation-based and probabilistic evidence. The root cause of this Grad-CAM failure lies in gradient saturation. Because the SegResNet model predicts this patient’s segmentation with extremely high confidence, the derivative of the sigmoid/softmax activation function approaches zero. This vanishes the gradient of the logit with respect to the bottleneck feature maps ($\partial \frac{y^c}{\theta} A^k \setminus \approx 0$), resulting in zero importance weights ($\alpha_k^c = 0$) and a blank saliency map. Because Occlusion Sensitivity relies on direct forward-pass perturbations rather than gradients, and GBP/IxG propagate gradients through different pathways, they are unaffected. This highlights gradient saturation as a primary vulnerability of gradient-based XAI in highly confident model states.

Second, the cross-method Weighted Dice rankings reveal method-dependent regional vulnerabilities. Occlusion Sensitivity leads for Whole Tumour (0.397) and Tumour Core (0.345), while Grad-CAM narrowly leads for Enhancing Tumour (0.250 vs. 0.233). This discrepancy is not a contradiction but a consequence of each method’s inductive bias: Occlusion captures empirical model dependency through physical perturbation, while Grad-CAM’s bottleneck weighting happens to better localise small ET structures when upsampled. Neither result alone tells the full story.

Third, the near-zero Saliency-Uncertainty Correlation (mean Pearson r across 23 patients: WT -0.007 , TC -0.037 , ET -0.095) confirms that saliency and uncertainty are independent signals. This independence is not a limitation but a design advantage: it means that jointly deploying both modalities provides a complete, non-redundant clinical interpretability layer. Saliency answers “Is the model attending to the right features?” while uncertainty answers “Where might the model be wrong?” — two fundamentally different clinical questions that require two fundamentally different instruments.

6.2. Clinical Implications

Three findings have direct implications for clinical deployment pathways. First, the high saliency coverage of Guided Grad-CAM (93% for WT across the evaluation cohort) indicates that the model concentrates its attention within the tumour boundary rather than on spurious image correlates — a necessary (though not sufficient) condition for clinical trust. This finding is strengthened by Occlusion Sensitivity’s independent confirmation: the highest Weighted Dice of all methods (0.397 WT) demonstrates that physically removing tumour voxels degrades the model’s predictions, providing empirical evidence that the model has learned tumour-relevant features rather than dataset-specific shortcuts.

Second, the boundary-concentrated uncertainty pattern observed across the 23-patient MC Dropout cohort (Boundary Uncertainty Ratios exceeding 0.84 for all regions) aligns with established neuro-oncological understanding. Tumour margins — particularly the infiltrative zone surrounding the enhancing rim — are inherently ambiguous on MRI due to partial volume effects and diffuse glioma cell infiltration. A well-calibrated model should exhibit elevated uncertainty precisely at these boundaries, and the observed pattern is consistent with this expectation.

Third, the positive Saliency-Uncertainty Correlation observed uniquely in patient 00291 (TC: $+0.090$, ET: $+0.053$) — where high feature relevance overlaps with high predictive variance — represents a specific pattern that, if validated on larger cohorts, could serve as an automated flag for cases requiring priority radiologist review. This signal is available only through multi-method analysis; no single XAI technique can simultaneously identify both high-relevance and high-uncertainty regions.

6.3. The Novel Weighted Dice Metric

The bottleneck resolution analysis reveals a critical insight: Saliency IoU’s volatile swings across resolutions make it unreliable for coarse-resolution methods like Grad-CAM. Our novel Weighted Dice metric provides stable evaluation by treating continuous saliency values as soft membership scores rather than forcing arbitrary binarization thresholds. While soft Dice formulations are commonly used as loss functions during training (e.g., [9]), their application as a post-hoc evaluation metric for continuous XAI saliency maps has not, to our knowledge, been previously explored.

This contribution is particularly important for 3D medical XAI, where saliency maps represent gradients of importance rather than binary decisions. Researchers evaluating coarse-resolution XAI methods should consider adopting Weighted Dice as a primary metric alongside existing measures.

6.4. Statistical Significance of XAI Rankings

To confirm that the differences in saliency alignment scores are not due to random variation, we conducted paired Wilcoxon signed-rank tests across the 25-patient evaluation cohort. For the Whole Tumour (WT) region, Occlusion Sensitivity’s superiority in Weighted Dice over all other methods is statistically significant: Grad-CAM ($p = 0.048$), Guided Backpropagation ($p < 0.001$), Guided Grad-CAM ($p < 0.001$), and Input $\times$ Gradient ($p < 0.001$). For the Tumour Core (TC) region, Occlusion’s performance is statistically significantly superior to GBP ($p < 0.001$), Guided Grad-CAM ($p < 0.001$), and IxG ($p < 0.001$), while the comparison with Grad-CAM shows no statistically significant difference ($p = 0.692$). These results confirm the statistical robustness of perturbation-based validation as a tool for clinical model verification.

6.5. Limitations

This work has five primary limitations that should be considered when interpreting the results.

Dataset generalisability. The framework is evaluated exclusively on BraTS 2023. Cross-institutional validation using datasets with different scanner protocols, field strengths, or patient demographics is essential to assess robustness under domain shift.

Single test split. All segmentation metrics derive from a single deterministic split (seed=42). While the consistency of improvement across two independent training scales (250-patient prototype and 1,251-patient full model) partially mitigates this concern, k-fold cross-validation would provide more robust performance estimates. The computational cost (24 hours per training run) made this infeasible within the project timeline.

XAI cohort size. The 25-patient XAI evaluation cohort, while substantially larger than the case-study approach common in prior work, represents approximately 20% of the test set. Extending quantitative XAI evaluation to the full 126-patient test set would strengthen the generalisability of cross-method conclusions, particularly for the Enhancing Tumour region where method performance is most variable.

Computational cost of Occlusion Sensitivity. Occlusion Sensitivity requires approximately 1,000 forward passes per patient, limiting its practicality for real-time clinical workflows. More efficient perturbation strategies (e.g., adaptive stride or region-of-interest-focused occlusion) should be explored.

Absence of clinical validation. Formal evaluation involving practising radiologists or neurosurgeons remains essential. The XAI metrics used in this work — Pointing Game, Weighted Dice, and others — are quantitative proxies for clinical utility. Whether these metrics correlate with diagnostic confidence or decision quality in practice is an open question that requires prospective clinical studies.

7. Conclusion

This paper presents a multi-method Explainable AI framework for 3D brain tumour segmentation that advances the integration of model accuracy and clinical interpretability. By deploying six complementary XAI techniques with a SegResNet architecture achieving competitive BraTS 2023 performance (Dice: 0.923 WT, 0.891 TC, 0.873

ET), and evaluating them quantitatively across a 25-patient cohort, this work establishes three key principles for clinical XAI deployment:

First, **multi-method evaluation is not optional** — it is a prerequisite for responsible clinical deployment. The gradient-based failure case demonstrates that silent method failure can produce misleading conclusions about model behaviour, a risk mitigated only through cross-paradigm redundancy.

Second, **saliency and uncertainty are complementary, not interchangeable**. Their statistical independence (Pearson $r \approx 0$) across the evaluation cohort confirms that deploying both modalities provides non-redundant clinical information — trust calibration through attribution, risk assessment through uncertainty.

Third, **evaluation metrics must match method resolution**. The novel Weighted Dice metric addresses the instability of hard-thresholded Saliency IoU for coarse-resolution methods, providing a principled foundation for cross-method comparison in 3D medical XAI.

These findings are subject to important limitations: reliance on a single dataset (BraTS 2023), the absence of formal clinical validation, and a single deterministic test split. Future work will prioritise formal evaluation with practising neuroradiologists, cross-institutional validation on diverse external datasets such as the Brain Tumour Segmentation (BraTS) historical cohorts (e.g., BraTS 2021) or the Clinical Proteomic Tumor Analysis Consortium (CPTAC) glioblastoma dataset, and extension of the XAI evaluation to the full 126-patient test set. Ultimately, this research demonstrates that bridging the trust gap in medical AI does not require sacrificing predictive accuracy, but rather demands a rigorous, multi-paradigm approach to explanation that matches the multi-dimensional complexity of clinical decision-making.

8. Appendix: Extended XAI Results

8.1. XAI Quantitative Performance: Full 4-Metric Comparison

The following tables show the complete quantitative evaluation of the five post-hoc saliency methods across the 25-patient cohort.

Method	Pointing Game	Saliency Coverage	Saliency IoU	Weighted Dice
Grad-CAM	92.0%	0.754 ± 0.242	0.149 ± 0.145	0.320 ± 0.145
GBP	64.0%	0.074 ± 0.088	0.010 ± 0.009	0.066 ± 0.051
Guided Grad-CAM	92.0%	0.806 ± 0.248	0.009 ± 0.019	0.132 ± 0.072
IxG (LRP Proxy)	92.0%	0.616 ± 0.262	0.004 ± 0.006	0.074 ± 0.033
Occlusion	100.0%	0.496 ± 0.198	0.257 ± 0.134	0.397 ± 0.116

Table 13: Whole Tumour (WT) region quantitative saliency comparison across the evaluation cohort (mean $\pm$ std).

Method	Pointing Game	Saliency Coverage	Saliency IoU	Weighted Dice
Grad-CAM	80.0%	0.549 ± 0.266	0.226 ± 0.183	0.358 ± 0.161
GBP	80.0%	0.079 ± 0.128	0.022 ± 0.016	0.053 ± 0.052
Guided Grad-CAM	92.0%	0.616 ± 0.256	0.012 ± 0.010	0.186 ± 0.082
IxG (LRP Proxy)	88.0%	0.511 ± 0.246	0.010 ± 0.011	0.122 ± 0.050
Occlusion	88.0%	0.317 ± 0.172	0.369 ± 0.155	0.345 ± 0.135

Table 14: Tumour Core (TC) region quantitative saliency comparison across the evaluation cohort (mean $\pm$ std).

Method	Pointing Game	Saliency Coverage	Saliency IoU	Weighted Dice
Grad-CAM	33.3%	0.282 ± 0.200	0.161 ± 0.175	0.250 ± 0.174
GBP	66.7%	0.031 ± 0.052	0.039 ± 0.074	0.041 ± 0.049
Guided Grad-CAM	75.0%	0.380 ± 0.206	0.017 ± 0.019	0.169 ± 0.073
IxG (LRP Proxy)	95.8%	0.387 ± 0.218	0.017 ± 0.021	0.141 ± 0.056
Occlusion	54.2%	0.177 ± 0.119	0.238 ± 0.122	0.233 ± 0.116

Table 15: Enhancing Tumour (ET) region quantitative saliency comparison across the evaluation cohort (mean $\pm$ std).

8.2. Supplementary Case Study: Patient 01518

To provide additional validation of XAI consistency, we report the complete saliency map set for patient 01518 in Figure 9.

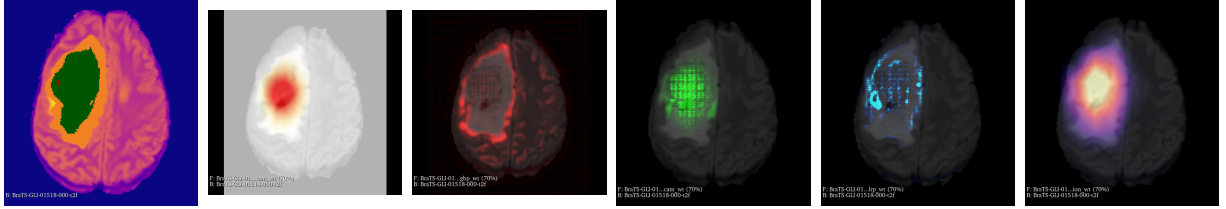


Figure 9: Comparison of saliency maps for patient 01518. From left: Ground Truth, Grad-CAM, Guided Backpropagation, Guided Grad-CAM, IxG, and Occlusion Sensitivity. Localisation remains highly consistent with WT/TC boundaries across methods, confirming model generalisability on this standard case.

8.3. VR Visualization Pipeline

All XAI outputs are exported as spatially-aligned NifTI volumes compatible with 3D Slicer and SlicerVR (Figure 10). The pipeline: (1) generates saliency maps in the same MONAI-preprocessed coordinate space as ground truth labels; (2) exports as NifTI format preserving spatial alignment; (3) enables direct loading into 3D Slicer for volumetric visualization; and (4) supports the SlicerVR extension for immersive exploration. This pipeline supports the transition from static 2D heatmaps to immersive 3D environments where clinicians can explore saliency and uncertainty distributions spatially, potentially reducing cognitive load inherent in traditional slice-by-slice interpretation.



Figure 10: VR pipeline output in SlicerVR. Left: Immersive volumetric rendering of patient brain MRI with tumour segmentation mask. Right: Interactive XAI saliency map overlay in VR space, allowing 6-DoF volumetric exploration of model attention.

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